

SAFETY DATA SHEET (SDS)

Included into the Register of Safety Data Sheets

RSDS № 3 7 4 1 5 3 3 9

from January 10, 2024

Expiration date to January 10, 2027

The Association «Non-commercial partnership «Coordination and Information Center of the CIS Member States for the Approximation of Regulatory Practices (CIC Center)

NAME

technical (по НД)

Universal waterproofing dry mixture D5 AQUASTOP

chemical (по IUPAC)

None

trade

Universal waterproofing dry mixture D5 AQUASTOP

synonyms

None

OKPD 2 Code

2 3 . 9 9 . 1 2 . 1 9 0

TN VED EAES Code

3 8 2 4 5 0 9 0 0 0

Symbol and name of the normative, technical or information document for the product
(GOST, TU, OST, STO, (M)SDS)

GPST P 56378-2015 Products and systems for the protection and repair of concrete structures.
Requirements for repair mixtures and adhesive joints of the contact zone when restoring structures

HAZARD CLASSIFICATION INFORMATION

Signal word

Danger

Brief (verbal description): Moderately hazardous product according to the degree of its impact on human body according to GOST 12.1.007. In case of skin absorption causes skin irritation. In case of eye absorption causes serious eye damage. May cause irritation of upper respiratory tract. In case of skin absorption may cause an allergic skin reaction. May cause damage to the respiratory system through repeated and prolonged exposure. May pollute the environment.

Detailed: in the 16 attached sections of the Safety Data Sheet

MAIN DANGEROUS COMPONENTS	MPC, mg/m ³	Hazard class	№ CAS	№ EC
Cement, portland	-/8	3	65997-15-1	266-043-4
Quartz	-/8	3	65997-15-1	266-043-4

APPLICANT SPE TOKAR Co., Ltd, Vladikavkaz
(name of the company) (town)

Type of applicant manufacturer, supplier, seller, exporter, importer
(cross out the unnecessary)

OKPO code 3 7 4 1 5 3 3 9

Emergency phone (8672) 51-17-73

Head of the applicant organization

(signature)

K. Bazoev /
(full name)



**Safety Data Sheet (SDS) complies with the UN recommendations
ST/SG/AC.10/30 «CFC (GHS)»**

IUPAC	– International Union of Pure and Applied Chemistry
GHS	– The UN recommendations ST/SG/AC.10/30 «Globally Harmonized System of Classification and Labelling of Chemicals (GHS)»
OKPD 2	– All-Russian Classifier of Products by Types of Economic Activity
OKPO	– All-Russian Classifier of Enterprises and Organizations
TN VED EAES	– Commodity Nomenclature of Foreign Economic Activity (CNFEA)
№ CAS	– Number of the substance in the Register «Chemical Abstracts Service» (CAS)
№ EC	– Number of the substance in the Register of «European Chemicals Agency» (ECHA)
MPC in the air of the working area	– Maximum permissible concentration of chemicals in the air of the working area, mg/m ³
Signal word	– a word used to emphasize chemical hazards and chosen according to GOST 31340-2013

1 Identification of chemical product and information about producer and/or supplier

1.1 Identification of chemical product

- 1.1.1 Technical name Universal waterproofing dry mixture D5 AQUASTOP [1]
- 1.1.2 Brief recommendations for use (including application restrictions) Universal waterproofing dry mixture D5 AQUASTOP is designed for repair and protection of concrete, reinforced concrete and stone structures [1].

1.2 Information about producer and/or supplier

- 1.2.1 Full legal name of the company Scientific and production enterprise TOKAR Co., Ltd
- 1.2.2 Address (postal and legal) 19 A Pozharskogo str., Vladikavkaz, 362002
- 1.2.3 Phone, including for emergency situations and time limits 8 (938) 883-10-37 from 9:00 a.m. to 18:00 p.m.
- 1.2.4 E-mail dobavkad5@mail.ru

2 Risk identification

- 2.1 Hazard classification of chemicals in general (information on hazard classification according to the law of the Russian Federation (GOST 12.1.007-76) and
- Moderately hazardous product according to the degree of its impact on human body (3 class of hazard according to GOST 12.1.007) [1-3].

SGS (GOST 32419-2013, GOST 32423-2013, GOST 32424-2013, GOST 32425-2013)

SGS classification:

- chemical products which cause damage (necrosis) / skin irritation: class 2;
- chemical products which have sensitizing effect when contacting skin;
- chemical products which cause serious damage / eye irritation: class 1;
- chemical products with selective toxicity to target organs and / or systems after a single exposure: class 3 (irritant effect);
- chemical products with selective toxicity to target organs and / or systems after a repeated / prolonged exposure: class 2 [4-7, 29].

2.2 Warning label information according to GOST 31340-2013

- 2.2.1 Signal word Danger [11].

- 2.2.2 Symbols (signs) of danger



- 2.2.3 Short characteristic of hazard (H-phrases)

- H315: Causes skin irritation.
H317: May cause an allergic skin reaction.
H318: Causes serious eye damage.
H335: May cause respiratory irritation.
H373: May cause damage to the respiratory system

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through repeated and prolonged exposure [1, 29].

3 Composition (components information)

3.1 The product information in general

3.1.1 Chemical name (according to IUPAC)	None [1]
3.1.2 Chemical formula	None [1]
3.1.3 General characteristic of the composition (considering brand assortment; method of production)	Dry mix is produced on the basis of high-strength cement, fine-grained quartz sand, the admixture D5, polypropylene fiber and special modified admixtures which increase adhesion and compensate for shrinkage [1].

3.2 Components

(name, numbers CAS and EC, mass fraction (the sum must be 100%), maximum permissible concentration (MPC) or tentative safe exposure level (TSEL), hazard classes, references to data sources)

Table 1 [3,8,29]

Components (name)	Mass fraction, %	Hygienic standards for workroom air		№ CAS	№ EC
		MPC, mg/m ³	Hazard class		
Cement, Portland	34,8	-/8 (a)	3 (F)	65997-15-1	266-043-4
Quartz	64	3/1 (a) (by silicon dioxide)	3 (F)	14808-60-7	238-878-4
Polypropylene	0,2	10 (a)	3	9003-07-0	618-352-4
Admixture D-5	1	3/1 (a) (by silicon dioxide)	3 (F)	No	No

Notes:

«a» - aerosol (predominant state of aggregation of a substance in the air under production conditions);
«F» - aerosols of predominantly fibrogenic action [9].

4 First aid procedures

4.1 Symptoms

4.1.1 In case of inhalation poisoning (by inhaling)	Cough, sore throat, dry mouth, irregular breathing [12].
4.1.2 In case of skin contact	Slight redness, slight swelling, itching [12].
4.1.3 In case of eye contact	Severe erythema, hyperemia, lacrimation, eyelid swelling, pain [12].
4.1.4 In case of oral poisoning (by swallowing)	Abdominal pain, nausea, vomiting [12].

4.2 First aid procedures

4.2.1 In case of inhalation poisoning	Fresh air, peace, warm. Seek medical attention, if necessary [12].
4.2.2 In case of skin contact	Rinse with running water. Seek medical attention, if necessary [12].

- | | |
|------------------------------|--|
| 4.2.3 In case of eye contact | Rinse with running water with wide open palpebral fissure for 15 minutes. Seek medical attention immediately [12]. |
| 4.2.4 In case of ingestion | Plentiful water drinking, activated charcoal, saline laxative. Seek medical attention, if necessary [12]. |
| 4.2.5 Contraindications | Don't give anything to an unconscious victim [12]. |

5 Fire and explosion prevention

- | | |
|---|---|
| 5.1 General characteristic of fire and explosion hazards
(according to GOST 12.1.044-89) | Non-combustible substance [1, 18] |
| 5.2 Indicators of fire and explosion hazards (nomenclature of indicators according to GOST 12.1.044-89) | Fire and explosion hazard indicators are not achieved [1, 19] |
| 5.3 Combustion and/or thermal degradation products and the hazard they cause | Not subject to combustion and thermal degradation [17, 30] |
| 5.4 Recommended fire-extinguishing measures | Extinguish the main source of ignition [1, 19] |
| 5.5 Prohibited fire-extinguishing measures | None [19]. |
| 5.6 Personal protective equipment (PPE) for firefighting (PPE for firefighters) | Special protective clothing for a firefighter, including combat clothing of a firefighter, special protective clothing against increased thermal effects, special protective clothing of an insulating type (breathing apparatus with compressed air, oxygen-insulating devices, etc.). Protective equipment for hands, feet and head (mittens, gloves, safety shoes, helmets) [1]. |
| 5.7 Fire-extinguishing specifics | Paper or polymer packaging may be involved in the combustion process [17]. |

6 Measures to prevent and eliminate accidents and emergencies and their consequences

6.1 Measures to prevent harmful effects on people, the environment, buildings, structures, etc. in accidents and emergency situations

- | | |
|--|--|
| 6.1.1 General action required in accidents and emergency situations | Isolate the danger zone within a radius of 50 m. Remove bystanders. Enter the danger zone in protective equipment. Take fire safety measures. Do not smoke. Eliminate sources of fire and sparks. Keep to the windward side. Provide first aid to the injured and send them for medical examination [1, 17]. |
| 6.1.2 Personal protective equipment (PPE) in case of emergencies (PPE for emergency teams) | Special protective clothing, small size industrial gas mask PFM-1 with a universal protective cartridge PZU, self-contained protective individual kit with forced supply of purified air to the breathing zone с принудительной подачей в зону дыхания очищенного воздуха. Butyl rubber gloves, special footwear [1, 17] |

6.2 Procedure for liquidation of accidents and emergency situations

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6.2.1 Procedure in case of leakage, spillage (including measures for their elimination and precautionary measures to protect the environment)

Do not touch the spilled substance. Spilled substance should be covered with sand or other inert material, collected in dry containers, and sent for disposal. Isolate the place where the product has spilled with sand, rinse abundantly with water and prevent the substance from entering the surface water. Indoors, cleaning work should be carried out with the ventilation turned on, using personal protective equipment [1, 17].

6.2.2 Procedure in case of fire

The product does not burn. Enter the accident place in protective clothing and breathing apparatus. Provide first aid to the injured [1, 17].

7 Storage of chemicals and their handling during loading and unloading operations

7.1 Safety measures when handling chemicals

7.1.1 Engineering Safety Systems

The technological process is maximally mechanized, the equipment is sealed. Production facilities must be equipped with a general combined extract-and-input and local ventilation system. The equipment must be sealed. It is necessary to comply with the fire safety rules. Workplaces must be equipped with the primary fire extinguishing means [1].

7.1.2 Environmental protection measures

Strict control and compliance with technological processes during use. Periodic control of the content of harmful substances in the air of the working area. Analysis of industrial effluents for the content of harmful substances in them in permissible concentrations. Purification of the air of industrial facilities to acceptable standards before being released into the atmosphere [1, 8].

7.1.3 Recommendations for safe transportation

The product can be transported by all types of transport in accordance with transportation rules for this type of vehicle.

To ensure the safety of the cargo, the cargo is enlarged by bundling the transport packaging [1].

7.2 Storage of chemicals

7.2.1 Conditions and terms of safe storage (including warranty period of storage; expiration date; substances and materials incompatible during storage)

The product is stored in original packaging in covered warehouses with relative humidity of the air of no more than 70% at a temperature of not lower than +10°C under conditions which can ensure the safety of packaging and protection from moisture. Store separately from acids and alkalis. Warranty period of storage of the product is 6 months from the date of production [1].

7.2.2 Containers and packing (including the materials they are made of)

In soft containers MKR with an insert with a capacity of 1000 kg, paper bags with an insert with a capacity of 25-30 kg [1]

7.2.3 Safety measures and storage at home

Not applicable at home [1]

8 Hazard control measures and personal protective equipment

8.1 Parameters of working area subject to mandatory control
(maximum permissible concentration (MPC) or tentative safe exposure level (TSEL) of the working area)

Cement, portland: $\text{MPC} = 8 \text{ mg/m}^3$, aerosol;
Quartz: $\text{MPC} = 3/1 \text{ mg/m}^3$, aerosol;
Polypropylene: $\text{MPC} = 10 \text{ mg/m}^3$, aerosol [1, 8].

8.2 Measures to ensure the content of harmful substances in permissible concentrations

Sealing of equipment, automation of technological operations, periodic monitoring of the air condition of the working area should be provided; premises for work should be provided with combined extract-and-input and local ventilation systems [1].

8.3 Personal protective equipment for personnel

8.3.1 General recommendations

Observe safety regulations, fire safety regulations and occupational hygiene regulations. In production area it is forbidden to eat, drink and smoke. At the end of work wash your face and hands with warm water and soap. It is necessary to undergo preliminary medical examinations when applying for a job and periodic examinations in accordance with the legislation of the Russian Federation.

Persons engaged in production and application of the product must be provided with protective clothing and personal protective equipment [1].

8.3.2 Respiratory protection (types of PPE)

Dust masks, universal respirators with filter cartridges of grade A [1, 31].

8.3.3 Protective equipment (material, type) (protective clothing, protective footwear, hand protection, eye protection)

Protective clothing of cotton, protective footwear, mittens and rubber gloves, safety glasses [1, 33].

8.3.4 Personal protective equipment for household use

Not applicable at home [1]

9 Physical and chemical properties

9.1 Physical state (aggregate state, color, odor)

Powder of grey color, odorless [1].

9.2 Parameters characterizing the main properties of the product (temperature indicators, pH, solubility, coefficient n-octanol/water and other parameters typical for this type of product)

Effective specific activity of natural radionuclides, Bq/kg, not more: 370;

Humidity of dry mixture, %, not more: 5;

Solubility: is not soluble in water, reacts with further hardening [1].

10 Stability and reactivity

10.1 Chemical stability (for unstable products specify decomposition products)

The product is stable under normal conditions of production, storage and transportation [29, 30].

10.2 Reactivity

Reacts with acids and alkalis [1, 30].

10.3 Conditions to avoid (including dangerous manifestations when contacting incompatible substances and materials)

Avoid contact with acids and alkalis. Avoid moisturizing the product during storage [1, 30].

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11 Toxicity information

11.1 General characteristic of effects
(hazard (toxicity) evaluation of impact on the
body and most typical manifestation of hazard)

Moderately hazardous product by the degree of impact on the body (3 class of hazard according to GOST 12.1.007). Causes irritation in case of skin contact. Causes irreversible effects in case of eye contact. May cause irritation of upper respiratory tract. May cause an allergic reaction in case of skin contact. May cause damage to the respiratory system through repeated and prolonged exposure [1, 12, 29].

11.2 Routes of exposure
(inhalation, ingestion, skin and eye absorption)

Inhalation, skin absorption, eye absorption, ingestion [12].

11.3 Affected organs, tissues and
human systems

Respiratory and nervous systems, bone tissue, gastrointestinal tract, morphological composition of peripheral blood, liver, kidneys [12]

11.4 Information about hazardous to
health effects in case of direct contact
with the product and consequences of
these effects (irritant effect on upper
respiratory tract, eyes, skin; skin-resorptive and
sensitizing effect)

Causes irritation in case of skin contact. Causes irreversible damage in case of eye contact. May cause irritation of upper respiratory tract.

Skin-resorptive effect is not established.

Has sensitizing effect in case of skin contact [12, 29, 32].

11.5 Information about hazardous long-
term effects of the product on the body
(effect on reproduction function,
carcinogenicity, mutagenicity, cumulative effect
and other chronic effects)

May cause damage to the respiratory system as a result of repeated or prolonged exposure, cement and quartz have a fibrogenic effect.

Portland cement has a reprotoxic effect (it is not classified according to GHS). Carcinogenic, mutagenic and teratogenic effects are not established.

For other components of the product carcinogenic, mutagenic, reprotoxic and teratogenic effects are not established.

Cumulative effect of the cement is moderate, for other components it is weak [29, 32].

11.6 Indicators of acute toxicity
(LD₅₀, routes of exposure (ingestion, skin
absorption), type of animal; LC₅₀, time of
exposure (h))

LD₅₀ > 2000 mg/kg, ingestion, rats (estimated time)

LC₅₀ > 1000 mg/m³, rats, 4 h (estimated time)

LD₅₀ > 2000 mg/kg, skin absorption, rats (estimated time)

Portland cement:

LD₅₀ > 2000 mg/kg, ingestion, rats

LC₅₀ > 1 mg/l of air, rats, 4 h

LD₅₀ > 2000 mg/kg, skin absorption, rats

Quartz:

LD₅₀ > 5000 mg/kg, ingestion, rats

LC₅₀ > 2,08 mg/l of air, rats, 4 h

LD₅₀ > 5000 mg/kg, skin absorption, rabbits

Polypropylene:

LD₅₀ > 110000 mg/kg, ingestion, rats

LD – lethal dose

LC – lethal concentration

For the admixture D-5 indicators have not been established [25, 29, 32].

12 Environmental impact information

12.1 General characteristic of the impact on the environmental objects (atmosphere, ponds, soils including the observed signs of impact)

The product can pollute the atmosphere, ponds and soils in case of violation of the rules of treatment. It can cause death of the aquatic life and can change organoleptic characteristics of water, can cause foaming, violation of processes of self-purification of ponds. It also leads to dustiness of vegetation cover and soil [1, 29].

12.2 Methods of environmental impact

Violation of storage rules and transportation of the product, unorganized disposal and incineration of waste, spillway into ponds, discharge into ponds and on land, accidents and emergencies [1]

12.3 The most important characteristics of the environmental impact

12.3.1 Hygienic standards (permissible concentrations in atmosphere, water, including fishery reservoirs and soils)

Table 2 [8, 23]

Components	Maximum permissible concentration (MPC) atm. or tentative safe exposure level (TSEL) atm., mg/m ³ (LHI ¹ , hazard class)	Maximum permissible concentration (MPC) water ² or approximate permissible level (APL) water, mg/l, (LHI, hazard class)	Maximum permissible concentration (MPC) fishery water ³ or tentative safe exposure level (TSEL) fishery water, mg/l (LHI, hazard class)	Maximum permissible concentration (MPC) of soil or tentative allowable concentration (TAC) of soil, mg/kg (LHI)
Cement, Portland	0,3/0,1, resorptive, 3 hazard class	10, san.-tox., 2 hazard class (silicon)	aluminum (for water-soluble forms) 0,04 mg/l, tox., 4 hazard class; iron (for water-soluble forms) 0,1 mg/l, tox., 4 hazard class; for marine water bodies 0,05 mg/l, tox., 2 hazard class; calcium (for water-soluble forms) 180 mg/l, san.-tox., 4 (ecological) hazard class; for marine water bodies 610 mg/l at 13-18%, tox., 4 (ecological) hazard class	Not known
Quartz	0,15/0,05, resorptive, 3 hazard class (inorganic dust, containing more than 70% SiO ₂)	10, san.-tox., 2 hazard class (silicon)	10,0, org., 3 hazard class (for sea water)	Not known
Polypropylene	0,1 tentative safe exposure level (TSEL) (polypropylene dust)	0,3, org. foam, 4 hazard class. (polypropylene emulsion)	Not known	Not known
Admixture D-5	Not known	Not known	Not known	Not known

12.3.2 Indicators of eco-toxicity (CL, EC, NOEC and others for fish (96 h.), daphnias (48 h.), water plants (72 or 96 h.) and

LC₅₀ > 100 mg/l, fish, 96 h (estimated time)

Quartz:

LC₅₀ > 100 mg/l, Danio rerio, 96 h

¹LHI – limiting harmful index (tox. – toxicological; san.-tox. – sanitary-toxicological; org. – organoleptic).

²Water of water objects of cultural and household water use

³Water of water objects of fishery importance (including sea water objects)

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others)

12.3.3 Migration and transformation in the environment through biodegradation and other processes (oxidation, hydrolysis etc.)

EC₅₀ > 1003 mg/l, Daphnia magna, 24 h.
Indicators for other components are not known [29]
After hardening the mix does not practically transform in the environment [29]

13 Recommendations for the waste disposal

13.1 Safety measures for handling waste generated during use, storage, transportation

Use PPE. Safety measures are the same as those recommended for the main product (see sections 7 and 8 of SDS) [1]

13.2 Information about places and methods of neutralization, recycling or waste disposal of the product, including containers (packing)

Dispose in places agreed with sanitary or environmental authorities in accordance with current sanitary standards and rules in places authorized by the local authorities of Rospotrebnadzor and Ministry of Natural Resources according to SanPin 2.1.3684-21. Bags from the product are sent for recycling or disposal [1, 24].

13.3 Recommendations for the disposal of waste generated during domestic use of the product

Not applicable at home [1]

14 Transportation information

14.1 Number of UN
(in accordance with UN recommendations on the transport of dangerous goods)

None [9]

14.2 Proper shipping and transport name

Proper shipping name: None

Proper transport name: Universal waterproofing dry mixture D5 AQUASTOP [1, 9]

14.3 Mode of transport used

Can be transported by all modes of transport in accordance with shipping rules for this mode of transport [9, 14].

14.4 Classification of dangerous goods according to GOST 19433-88:

- class
- sub-division
- classification code (according to GOST 19433-88 and during rail transportation)
- number(s) of drawing(s) of hazard signs

Is not classified as dangerous goods [14].

None [14]

None [14, 16]

None [14]

14.5 Classification of dangerous goods according to the UN Recommendations on the Transport of Dangerous Goods:

- class or sub-division
- subsidiary hazard class
- the UN packing group

Is not classified as dangerous goods [9]

None [9]

None [9]

14.6 Transport marking (manipulation symbols according to GOST 14192-96)



- «Keep dry» [1, 16]

14.7 Transport Emergency Cards
(for transportation by rail, sea and others)

Not applicable [17, 21, 22]

15 Information on national and international legislation

15.1 National legislation

15.1.1 Legislation of the Russian Federation

Federal Law №184-FZ of December 27, 2002 «On Technical Regulation» (with the Amendments and Additions of July 2, 2021);

Federal Law №52-FZ of March 30, 1999 «On Sanitary and Epidemiological Welfare of the Population» (with the Amendments of July 2, 2021);

Federal Law №7-FZ of January 10, 2002 «On Environmental Protection» (with the Amendments of July 2, 2021);

Federal Law №116-FZ of July 21, 1997 «On Industrial Safety of Hazardous Production Facilities» (with the Amendments of June 11, 2021);

Federal Law №89-FZ of June 24, 1998 «On Production and Consumption Waste» (with the Amendments of July 2, 2021);

Federal Law №96-FZ of May 04, 1999 «On the Protection of the Atmospheric Air» (with the Amendments of June 11, 2021);

Federal Law №123-FZ of July 22, 2008 «Technical Regulations on Fire Safety Requirements» (with the Amendments of April 30, 2021);

Not required [35].

15.1.2 Information about the documentation regulating the requirements for the protection of the human and the environment

15.2 International conventions and agreements (whether the product is regulated by the Montreal Protocol or the Stockholm convention and others)

Not regulated by the Montreal Protocol and the Stockholm Convention [33, 34]

16 Additional information

16.1 Information about the revision (reissue) of the safety data sheet (SDS)

SDS is created for the first time

16.2 List of data sources used in compiling the Safety Data Sheet³

1. GOST P 56378-2015 Products and systems for the protection and repair of concrete structures. Requirements for repair mixtures and adhesive joints of the contact zone when restoring structures (with the Amendment №1)
2. GOST 30333-2007. Safety Data Sheet of chemical products
3. GOST 12.1.007-76 CCBT. Hazardous substances. Classification and general safety requirements (with the Amendments №1, 2)
4. GOST 32419-2013 Hazard classification of chemical products. General requirements
5. GOST 32423-2013 Hazard classification of mixed chemical products by their effects on the body

³ The serial numbers of data sources are given in each paragraph of the SDS in the form of links

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6. GOST 32424-2013 Hazard classification of chemical products by environmental impact. Key points
7. GOST 32425-2013 Hazard classification of mixed chemical products by environmental impact
8. SanPin 1.2.3685-21. Hygienic Standards and Requirements for Ensuring Safety and (or) Harmlessness of Environmental Factors for Humans
9. The UN Recommendations on the Transport of Dangerous Goods «Orange Book»: Model Regulations – Twenty-second Revised Edition. The UN, 2021
10. The Information system data GESTIS Substance Database. [Electronic resource]: Access mode - <https://gestis-database.dguv.de/>
11. GOST 31340-2013 Warning labelling of chemicals. General requirements
12. The Russian register of potentially hazardous chemical and biological substances. [Electronic resource]: Access mode - <http://www.rpohv.ru/>
13. GOST 12.1.005-88 Occupational Safety Standards System (CCBT). General sanitary requirements for working zone air (with the Amendment №1)
14. GOST 19433-88. Dangerous goods. Classification and marking (with the Amendment №1)
15. GOST 12.4.034-2017. CCBT. Respiratory protective equipment. Classification and marking
16. GOST 14192-96. Marking of cargoes (with the Amendments №1, 2, 3)
17. Emergency cards for dangerous goods transported by railways of the CIS, the Republic of Latvia, the Republic of Lithuania, the Republic of Estonia (with the Amendments of November 27, 2020) [Text]: approved by the Commonwealth Member State Rail Transport Council, Protocol №48 of May 30, 2008
18. GOST 12.1.044-89 Occupational Safety Standards System (CCBT). Fire and explosion hazard of substances and materials. Nomenclature of indicators and methods of their determination
19. A.Y. Korolchenko, D.A. Korolchenko. Fire and explosion hazard of substances and materials and means of extinguishing. Guide: in 2 parts. – 2-nd edition, revised and updated. – M.: Ass. «Pozhnauka», 2004
20. On approval of Rules of cargo transportation by motor transport and about amendments to Clause 2.1.1 of the Traffic Rules of the Russian Federation [Text]: Government Decree №2200 of December 21, 2020
21. Data of UPS Chemical Table □ ICAO/IATA. [Electronic resource]: Access mode - <https://www.ups.com>
22. The International Maritime Dangerous Goods Code (IMDG Code), volume 1. Spb.: ZAO TSNIIMF, 2007
23. On approval of water quality standards for water bodies of fishery significance, including standards for maximum permissible concentrations of harmful substances in the waters of water bodies of fishery significance (with the Amendment of March 10, 2020) [Text]: Order №552 of December 13, 2016 of the Ministry of Agriculture of the Russian Federation // The Russian Federation Code, 2008, №27, the Art. 3286; 2012, №44, the Art. 6026
24. SanPin 2.1.3684-21. Sanitary and epidemiological requirements for maintenance of the territories of urban and rural settlements, for water bodies, potable water and potable water supply, air, soils, living quarters, operation of industrial, public facilities, organization and implementation of sanitary and anti-epidemic (preventive) measures
25. The Information system Merck KGaA data. [Electronic resource]: Access mode - <https://www.sigmaaldrich.com/>
26. GOST 12.4.021-75 Occupational Safety Standards System (CCBT). Ventilation systems. General requirements (with the Amendment №1)
27. SP 2.1.7.1386-03. Sanitary rules for determining the hazard class of toxic production and consumption wastes
28. European Agreement Concerning the International Carriage of Dangerous Goods by Road (ADR) (restructured ADR operating since January 1, 2019). – The UN, 2019

29. The Information system ECHA (European Chemicals Agency) data. [Electronic resource]: Access mode - <http://echa.europa.eu/>
30. Chemical Encyclopedia / Editorial board: Knunyants I.L. (chief ed.) and others. – M.: Sov. encycl., 1990
31. Collective and personal protection equipment. Control of protection equipment. Encyclopedia «Ecometry» from a series of reference books on environmental and medical measurements. - M.: FID «Delovoy ekspres», 2002.
32. The Information system ChemIDplus data [Electronic resource]: Access mode - <https://chem.nlm.nih.gov/chemidplus/chemidlite.jsp>
33. The Montreal Protocol on Substances that Deplete the Ozone Layer. – The UN, 1989
34. The Stockholm Convention on Persistent Organic Pollutants. – The UN, 2001
35. The Single List of Goods, subject to sanitary-and-epidemiologic supervision (control) on the customs border and customs area of the Customs union, approved by Decision №299 of the Customs Union Commission dated May 28, 2010 (as amended by Decisions of the Customs Union Commission №341 of 17.08.2010, №383 of 20.09.2010, №432 of 14.10.2010)